

# Real File Compression and Decompression System

Using C++ and gzip Compression Utility

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# Problem and Objective

## The Problem

- Large files consume significant storage space
- Slow data transfer without compression
- Simple algorithms like RLE fail on real-world files

## Project Goals

- Build file compression system using C++
- Compress any file type (text, PDF, images, binaries)
- Restore files exactly without data loss
- Demonstrate OOP concepts in practice



# System Architecture



User File

C++ Compressor

gzip Engine

Compressed File

# C++ Concepts Demonstrated



## Abstract Classes

Pure virtual functions for interface definition



## Inheritance

Runtime polymorphism with derived classes



## Encapsulation

Private members for data protection



## Exception Handling

Try/catch blocks for error management



## File I/O

Reading and comparing files efficiently



## System Calls

Invoking gzip compression utility

# Results and Conclusion

27%

Size Reduction

Bytes saved through compression

100%

Data Accuracy

Lossless decompression verified

## Key Achievements

- Successfully compresses real files using gzip
- Supports multiple file types (text, PDF, images)
- Compression ratio varies by file type
- Exact file restoration with verification

## Project Success

Demonstrates real-world file compression system implementing major C++ OOP concepts with lossless compression through file verification.